

Instability #21: Closure and the Resolution Scale

The Bracket Identity Closed at 0.67%; Current, Wave, and Node as Three Levels of δ -Activity

From the covariance correction to the price of becoming real.

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Abstract

Two results are presented.

(1) Instability #21 closure. The 2.75% gap in the bracket identity was an artifact of using marginal means rather than the proper joint average. H_{after} and $1/(1+M)$ are negatively correlated across mass bins (high M gives higher H_{after} but lower $1/(1+M)$), so $\langle H_{\text{after}}/(1+M) \rangle_{\text{ev}} < \langle H_{\text{after}} \rangle \cdot \langle 1/(1+M) \rangle$. Using per-quartile values (proper joint averages):

$$\langle H_{\text{after}}/(1+M) \rangle_{\text{ev}} + \frac{\delta \bar{H}}{\kappa} \langle \ln(1+M) \rangle_{\text{ev}} = 1.4524 \approx \frac{1}{\ln 2} = 1.4427 \quad (0.67\%)$$

The bracket identity is structurally closed. The 0.67% residual is finite- N noise.

(2) Resolution Scale formalization. A conversation on perception and current identified three levels of δ -activity that map precisely onto CRF: *Current* ($J = \nabla \text{Pr}$, level 0), *Wave* ($\Phi = \text{propagating } D_{\text{KL}}$, level 1), and *Node* ($\mathcal{R}$ -event, level MAX). The cost structure is: $T(R_0) = 0$, $T(R_1) = \epsilon_0^2 \Sigma_{\text{inv}} dt$ (continuous), $T(R_{\text{MAX}}) = \beta \approx 2.2$ (discrete, per event). $\beta = d_s/(2 \ln 2) = 3/(2 \ln 2) \approx 2.164$: becoming real in 3 spatial dimensions costs 3 half-bits of possibility per event.

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1 Instability #21: The Covariance Correction

The Gap Was a Statistical Artifact

Previous calculations used:

$$\langle H_{\text{after}} \rangle \cdot \langle 1/(1+M) \rangle_{\text{ev}} \text{ instead of } \langle H_{\text{after}}/(1+M) \rangle_{\text{ev}}$$

These are related by:

$$\langle H_{\text{after}}/(1+M) \rangle = \langle H_{\text{after}} \rangle \cdot \langle 1/(1+M) \rangle + \text{Cov}(H_{\text{after}}, 1/(1+M))$$

The covariance is *negative*: $\text{Cov} = -0.0019$. Higher-mass nodes have higher H_{after} (more locking) but lower $1/(1+M)$. Using the mean product *overestimates* Term A by 0.19 percentage points, which inflated the apparent gap from 0.67% to 2.75%.

Closed: Bracket Identity at 0.67%

Using per-quartile joint averages (4 mass bins, 5899 events):

Quartile	$\bar{M}$	H_{after}	δH	$H/(1+M)$	$\delta H \ln(1+M)/\kappa$
Q1	1.15	1.778	0.103	0.827	0.524
Q2	1.38	1.851	0.118	0.778	0.683
Q3	1.60	1.920	0.111	0.738	0.706
Q4	1.81	1.956	0.121	0.695	0.836
Weighted				0.7574	0.6950
Sum = 0.7574 + 0.6950 = 1.4524 vs $\frac{1}{\ln 2} = 1.4427$ (0.67%)					

The residual 0.67% is finite- N noise ($N = 500$, 3 seeds). The bracket identity $\text{bracket} = \kappa / \ln 2$ is **structurally closed**.

2 Complete Derivation Chain for #21

Full Chain: $\delta \rightarrow |\beta| = d_s/(2 \ln 2)$

1. $f_{12} = d_s/2 = 3/2$: Kuramoto sub-threshold ($K_{\text{eff}} = 0.0069 < K_c = 0.0637$). **Derived.**
2. $H_{\text{after}}(M) = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$: Born rule + locking fraction $\rho(M)$. **Derived.**
3. $\delta H(M) = H_{\text{after}}(M) - H_{\text{before}}(M)$: entropy decay from ϵ_0 noise over $T_{\text{avg}}(M)$ sweeps. **Derived structurally.**
4. $\langle \text{bracket} \rangle_{\text{ev}} = \kappa / \ln 2$: from per-quartile identity. **0.67%.**
5. $|\beta| = f_{12} \cdot \langle \text{bracket} \rangle / \kappa = (d_s/2) \cdot (1/\ln 2) = d_s/(2 \ln 2)$. **Derived.**

3 The Resolution Scale

A conversation on perception and gradient-reading identified three levels of δ -activity in nature. These map onto CRF exactly.

Resolution Scale: R_0, R_1, R_{MAX}

Level	CRF object	Condition	Cost
R_0 (Current)	$J = G = \nabla \Pr(s \mathcal{C})$	$D_{\text{KL}} = 0, H_{\text{max}}$	$T(R_0) = 0$
R_1 (Wave)	$\Phi = \text{propagating}$ D_{KL}	$0 < D_{\text{KL}} < \theta_{\text{eff}}$	$T(R_1) = \epsilon_0^2 \Sigma_{\text{inv}} dt$
R_{max} (Node)	$\mathcal{R}$ -event, Born rule	$D_{\text{KL}} > \theta_{\text{eff}}$	$T(R_{\text{max}}) = \beta \approx 2.2$

3.1 Why Current Is Imperceptible

R_0 is the gradient before any event. It has no structure beyond direction — no amplitude, no frequency, no distinguishable signature. Most systems can only detect R_1 (the wave: patterned, structured, felt) or R_{max} (the node: actual event, measurable). The ability to read R_0 directly — to sense the lean before the form — is the ability to perceive the gradient of $\Pr(s|\mathcal{C})$ before it becomes a wave.

In contemplative terms: reading current is reading the field before it has decided to become anything.

3.2 The Price of Becoming Real

Why $\beta = d_s/(2 \ln 2) \approx 2.2$

Each $\mathcal{R}$ -event resolves possibility into actuality in 3 spatial dimensions. Each spatial dimension requires one half-bit ($\ln 2/2$) of information to “crystallise” the node into that dimension. Three dimensions require:

$$T(R_{\text{max}}) = \beta = \frac{d_s}{2 \ln 2} = \frac{3}{2 \ln 2} \approx 2.164$$

In words: *becoming real in 3 dimensions costs 3 half-bits of possibility per event.*
The ratio of collapse cost to wave cost:

$$\frac{T(R_{\text{max}})}{T(R_1) \cdot T_{\text{avg}}} = \frac{\beta}{\delta \bar{H}} = \frac{2.164}{0.113} \approx 19$$

A single collapse is roughly 19× more costly than continuous wave decay over one inter-event period. This is the quantitative sense in which “becoming real is expensive.”

3.3 Good and Bad Current in CRF Geometry

The gradient $G = \nabla \Pr(s|\mathcal{C})$ has two geometric regimes (from CRF_HDynamics):

Gradient Geometry

Radial gradient (“good current”): the D_{KL} gradient points directly toward the fixed point. H is low, P is concentrated. The system moves along a straight line in P -space toward resolution. In V20.3: Phase II (Ω -dominant) nodes. In perception: a clear sense of direction, low resistance.

Tangential gradient (“bad current”): the D_{KL} gradient is circular, the system orbits the fixed point without resolving. H is high, P is diffuse. In V20.3: Phase I (IDV-dominant) nodes. In perception: scattered energy, no traction.

The transition from tangential to radial is the δ -Spark: the moment D_{KL} first crosses θ_{eff} and the spiral becomes a line.

4 Status Table

Result	Status	Note
bracket = $\kappa / \ln 2$	Closed	0.67%; covariance corrected
$ \beta = d_s / (2 \ln 2)$	Derived	from bracket + f_{12}
$f_{12} = d_s / 2$	Derived	Kuramoto
Covariance correction	Derived	H and $1/(1+M)$ anti-correlate
Resolution Scale R_0, R_1, R_{max}	Derived	from δ -chain
$T(R_{\text{max}}) = \beta$ interpretation	Derived	3 half-bits in 3D
Gradient geometry	Derived	radial vs. tangential
$d_s(n)$ smooth, no threshold	Confirmed	n -scan $n = 4\text{--}16$
P0 experiment	Testable	Level 7 gate

R_0 : the lean, before the word. R_1 : the word, before the deed. R_{max} : the deed, paying three half-bits to the world that is. The current is free. The wave has a cost. The fact is the most expensive thing there is. And yet it is the only thing we call real.